Network Principles Group Assignment

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Subject	NET1014 – NETWORK PRINCIPLE
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Work Breakdown Structure:

<table>
<tr><td colspan="5">CLO2</td></tr>
<tr><td colspan="5">Introduction
- Objective
- Assumptions (number of hosts, rooms, labs, etc.)</td></tr>
<tr><td rowspan="2">Individual Component</td><td>Student 1
Kuala Lumpur
(1st floor)</td><td>Student 2
Kuala Lumpur
(2nd floor)</td><td>Student 3
Cyber Jaya
(1st floor)</td><td>Student 4
Cyber Jaya
(2nd floor)</td></tr>
<tr><td>Floor plan-
(Layout)
Justification

Network diagram and configuration (Packet Tracer)</td><td>Floor plan-
(Layout)
Justification

Network diagram and configuration (Packet Tracer)</td><td>Floor plan-
(Layout)
Justification

Network diagram and configuration (Packet Tracer)</td><td>Floor plan-
(Layout)
Justification

Network diagram and configuration (Packet Tracer)</td></tr>
<tr><td></td><td colspan="4">Configuration/Demonstration</td></tr>
<tr><td colspan="5">CLO2</td></tr>
<tr><td>Group Component</td><td colspan="2">Kuala Lumpur
(1st and 2nd Floor)

Topology and IP Addressing Justification</td><td colspan="2">Cyber Jaya
(1st and 2nd Floor)

Topology and IP Addressing Justification</td></tr>
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1.0 Introduction

VanTech, a leading enterprise software company based in Kuala Lumpur, Malaysia, is renowned for offering a diverse range of products in both the enterprise and business-to-business SaaS sectors. VanTech is a technology partner for many global enterprises. The CEO intends to relocate the technical and development departments to Cyberjaya by mid-2023 while restructuring the existing network infrastructure in the Kuala Lumpur building and laying down a new network infrastructure. Both buildings, with two floors each, have distinct departmental allocations.

The primary aim of this assignment is to establish a network environment characterized by robustness, scalability, and security, facilitating smooth communication, data transfer, and collaboration between the two branches. To construct a comprehensive network infrastructure for the company, the identification of specific network requirements and the evaluation of strengths and weaknesses within each area are imperative. This process involves gaining an understanding of the unique needs of each department, ensuring that the resulting network configuration is finely tuned to enhance overall efficiency and meet the diverse demands of the organization.

VanTech, a software company catering to the enterprise sector and business-to-business SaaS space, prioritizes the implementation of robust security measures to protect sensitive data and resources. This includes access controls and authentication mechanisms to restrict entry to authorized personnel, ultimately bolstering network security. Considering the company's international presence, seamless network connectivity is vital for maintaining work efficiency. Achieving comprehensive network coverage throughout the building facilitates the use of personal devices, enhancing overall productivity. Additionally, network segmentation is recommended to isolate sensitive departments, preventing a security breach in one area from compromising the entire network.

The **Kuala Lumpur branch**, **first floor** has a reception area, waiting room, cafeteria, and Internet and CCTV monitor room. The **second floor** has a CEO room, KL branch manager room, Finance department, Human resource department, and administrative office room for business. In **Cyber Jaya branch, first floor** has a reception area, pantry for staff, cyber jaya branch manager room, and CTO room. The **second floor** has a development hub, Server room, technical assistant to monitor the network, and the meeting/conference room.

In the network diagram, if there are too many devices, a single device may be used to represent all of them and only one device is configured. Our team has been appointed to assist with the proposal to plan for a suitable network design for this new project. In this assignment, we will discuss the network diagram designed for Vantech, including the topology employed and the calculation of IP addresses, among other aspects. Also, the network address used in this assignment is 192.168.1.0 ~ 192.168.5.0.

2.0 Network topology

2.1 Star Topology

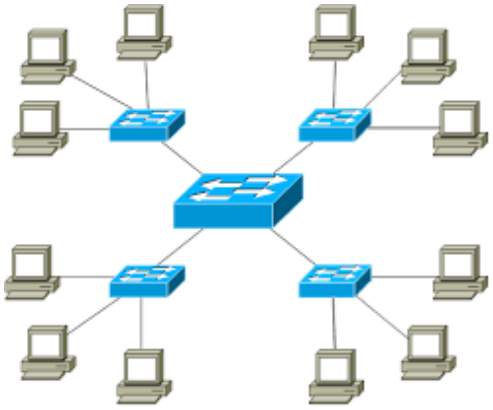


Figure 2.0 *Star topology*

A star topology is a network configuration where every network component is physically linked to a central node, such as a router, hub, or switch. This central node functions as a mediator for data transmission, facilitating communication between devices. In star topology, the central hub assumes a server-like role, while the connecting nodes serve as clients. When a packet is received from a connecting node, the central node can relay it to other nodes in the network.

Advantages

High Reliability	In a star topology, each connected node operates independently. If one node experiences a failure, it does not impact the performance of other nodes in the network.
Optimal Performance	The star topology minimizes the risk of data collisions as each device is directly linked to the central node through cables.
Centralized Network Monitoring	The network is efficiently monitored through a centralized management system utilizing the hub, central computer, or switch.

Easy Installation and Expansion	Adding new nodes to the star topology is easy and doesn't disrupt the existing network. The removal and replacement of components are also convenient, facilitating network maintenance. With the help of star topology, we can easily extend the length of the network right and progress on our from multiple stars in the middle with their central core as a server.
Manageability	Fault detection in the star topology is easily manageable, thanks to easily identifiable links. Troubleshooting and detecting failures within this topology are straightforward processes.

Table 2.0 *Advantages of Star Topology*

Disadvantages

Elevated Costs	Installation costs may rise, especially when using a router or a switch as the central network device.
Dependency on Central Hub	In star topology, if the central hub malfunctions, the entire system is affected as no devices can operate independently without it.
Intensive Maintenance	The central hub in a star topology demands regular maintenance and additional resources due to its pivotal role in the network.
Limited Data Transfer Rates in Wireless Systems	Compared to wired systems, wireless star topologies, particularly Wireless LANs (WLANs), might encounter reduced data transfer rates, increasing the likelihood of data bottlenecks.

Table 2.1 *Disadvantages of Star Topology*

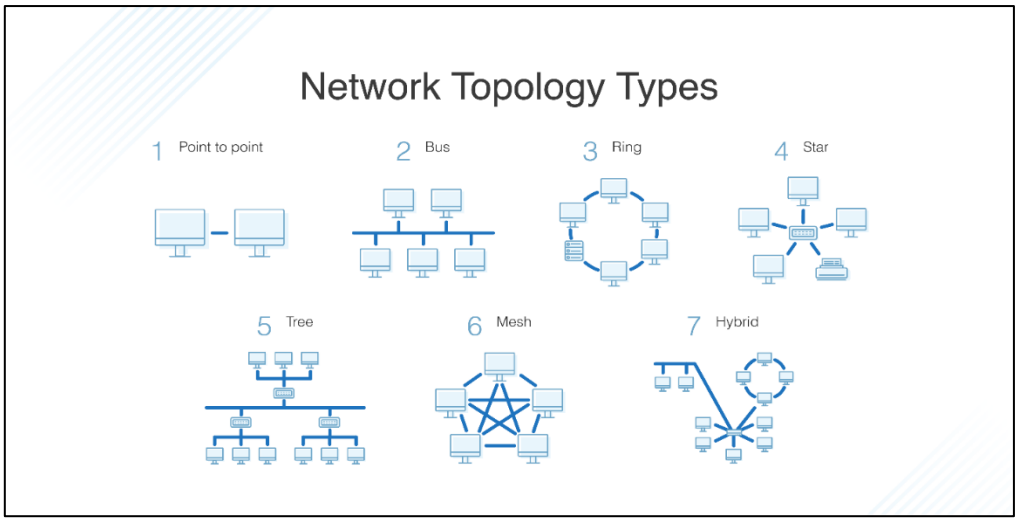


Figure 2.1 *Network Topology*

2.2 Bus Topology

Each workstation is linked in series to the main central cable in a bus topology. Small networks benefit greatly from their cost-effectiveness and straightforward linear design. Bus topology, however, is slow for larger networks and makes it hard to identify issues in the event of a network failure.

2.3 Ring Topology

In a ring topology, network devices are connected via cables such that the final device is linked to the initial device. Every device forms a continuous ring with exactly two other devices connected to it. Ring topology minimizes the possibility of packet collision and is economical. Ring topology, however, requires a single cable, is costly to maintain, and is challenging to troubleshoot.

2.4 Tree Topology

Tree topology is a network design that combines features of star and bus topologies. It has a hierarchical structure resembling a tree, with a central hub connecting to secondary hubs or switches, which then link to end devices. This arrangement allows for scalability and efficient data flow. While providing reliability and ease of expansion, the dependency on the central hub poses a vulnerability to network failures.

2.5 Mesh Topology

In this architecture, nodes are connected to one another so that there are paths available between a minimum of some network nodes. A network mapping that is "fully meshed" consists of all nodes connected to every other node, while a network that is "partially meshed" will only have some nodes connected to multiple nodes. Several paths are merged to increase network resilience. Dedicated links, however, require more space and come at a high cost.

2.6 Hybrid Topology

A network topology is referred to as hybrid when it combines two or more topologies in any way. Within a network, hybrid topology provides flexibility to various departments of an organization. A department may choose to use a specially designed network topology mapping that better fits its needs.

3.0 IP Address

3.1 Introduce to Network

Networking involves the interconnection of devices, such as computers and servers, to share resources and information. It enables communication and data exchange between these devices, facilitating efficient information flow. IPv4, or Internet Protocol version 4, is the fourth version of the Internet Protocol that assigns a unique numerical label, known as an IP address, to each device participating in a computer network. The structure of an IPv4 address consists of 32 bits, typically expressed as four octets separated by dots (e.g., 192.168.1.1). Each octet represents 8 bits, allowing for a total of 256 possible values (2^8).

IPv4 addresses are divided into two parts: the network portion and the host portion. The division between these two parts is determined by the subnet mask. The subnet mask specifies how many bits are used for the network and how many for the host.

IPv4 also follows certain rules:

- Each IPv4 address on a network must be unique.
- IPv4 addresses are 32-bit numerical labels often represented in dotted-decimal format.
- Dividing IP networks into sub-networks enhances performance and security.
- Specific addresses (network, broadcast, etc.) serve designated purposes.
- IP addresses can be assigned dynamically (DHCP) or statically.
- IP addresses are categorized into classes based on the range of the first octet and the number of networks and hosts they can support.

Class	First Octet Range	Default Subnet Mask	Prefix	Number of Networks	Number of Hosts
Class A	1-126	255.0.0.0	/8	125	16777214
Class B	128-191	255.255.0.0	/16	16382	65534
Class C	192-223	255.255.255.0	/24	2097150	254
Class D	224-239	Multicast addresses			
Class E	240-255	Restricted/Experimental			

Table 3.0 *IPv4 Classes*

3.2 FLSM and VLSM

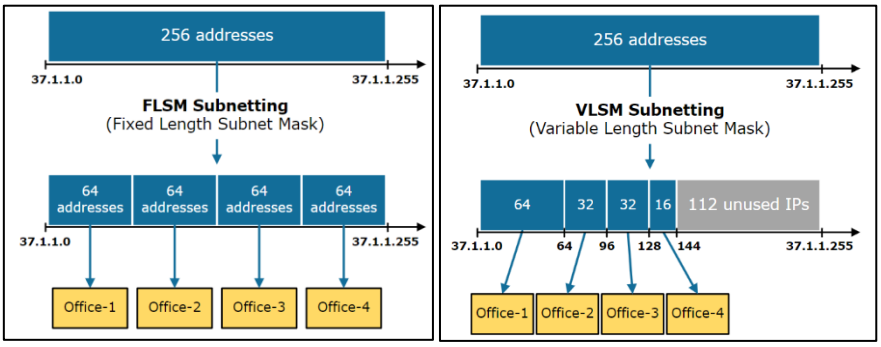


Figure 3.0 *FLSM and VLSM*

There are two techniques used for subnet an IP address for a network based on IPv4, which is **fixed-length subnet mask (FLSM)** and **variable-length subnet mask (VLSM)**. All subnets in a network based on FLSM will use the same subnet mask length. Meanwhile, different subnets may have different subnet mask lengths for VLSM. The following table shows the difference between FLSM and VLSM.

FLSM	VLSM

One network will divide into multiple same sized subnets.	One network will divide into different sized subnets.
Each subnet has an equal number of hosts.	Each subnet may have different number of hosts
The same subnet mask used for all subnets.	Different subnet mask use for each subnet.
Configuration and administration are straight forward.	Configuration and administration are more difficult.
Significance waste on IP address.	IP address waste minimized.

Table 3.1 *Differences of FLSM and VLSM*

4.0 Wide Area Network (WAN) of VanTech

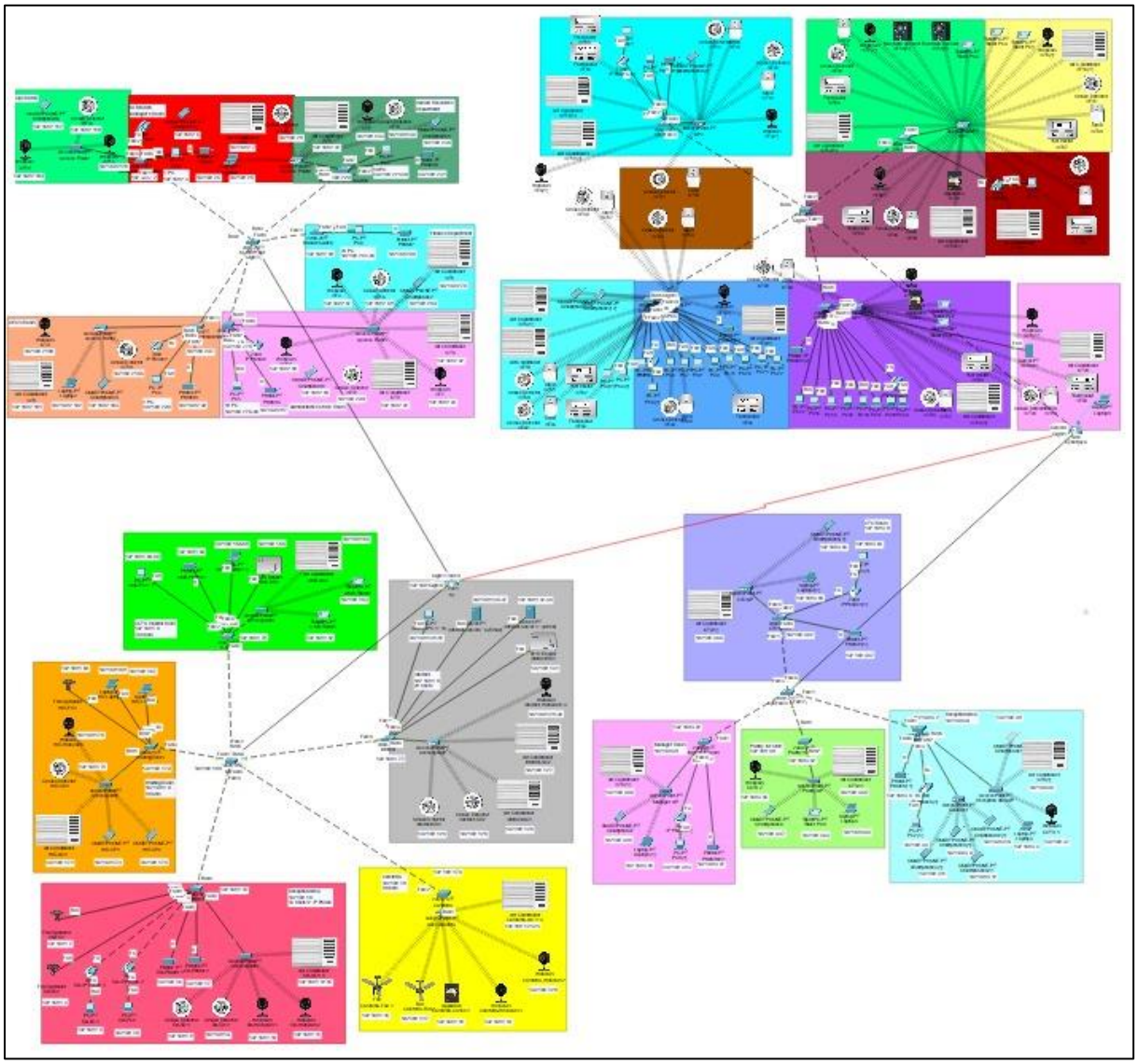


Figure 4.0 *VanTech Network Diagram*

Figure 4.0 shows the wide area network (WAN) of VanTech Company, which is the combination of 4 different floors in 2 branches, which are Kuala Lumpur and Cyberjaya. The left side of the diagram is from the Kuala Lumpur branch, and the right side is from the Cyberjaya branch. Table 4.0 shows the 5-network included in the WAN of the company. The KL branch and Cyberjaya branch each have a router placed in their respective Internet rooms, connected via a serial cable. Vantech has a total of three servers; the Email server and Web

Server are situated in the Internet room on the first floor of the KL branch, while the DNS server is located in the Internet room on the second floor of the Cyberjaya building. In the network diagram, if there are too many devices, a single device may be used to represent all of them and only one device is configured.

Area	Network Address
Kuala Lumpur 1st Floor	192.168.1.0
Kuala Lumpur 2nd Floor	192.168.2.0
Cyberjaya 1st Floor	192.168.3.0
Cyberjaya 2nd Floor	192.168.4.0
Between Routers	192.168.5.0

Table 4.0 *Network Address Table*

5.0 Kuala Lumpur First Floor

Lim Jia Xuan 2110787

5.1 Topology with justification

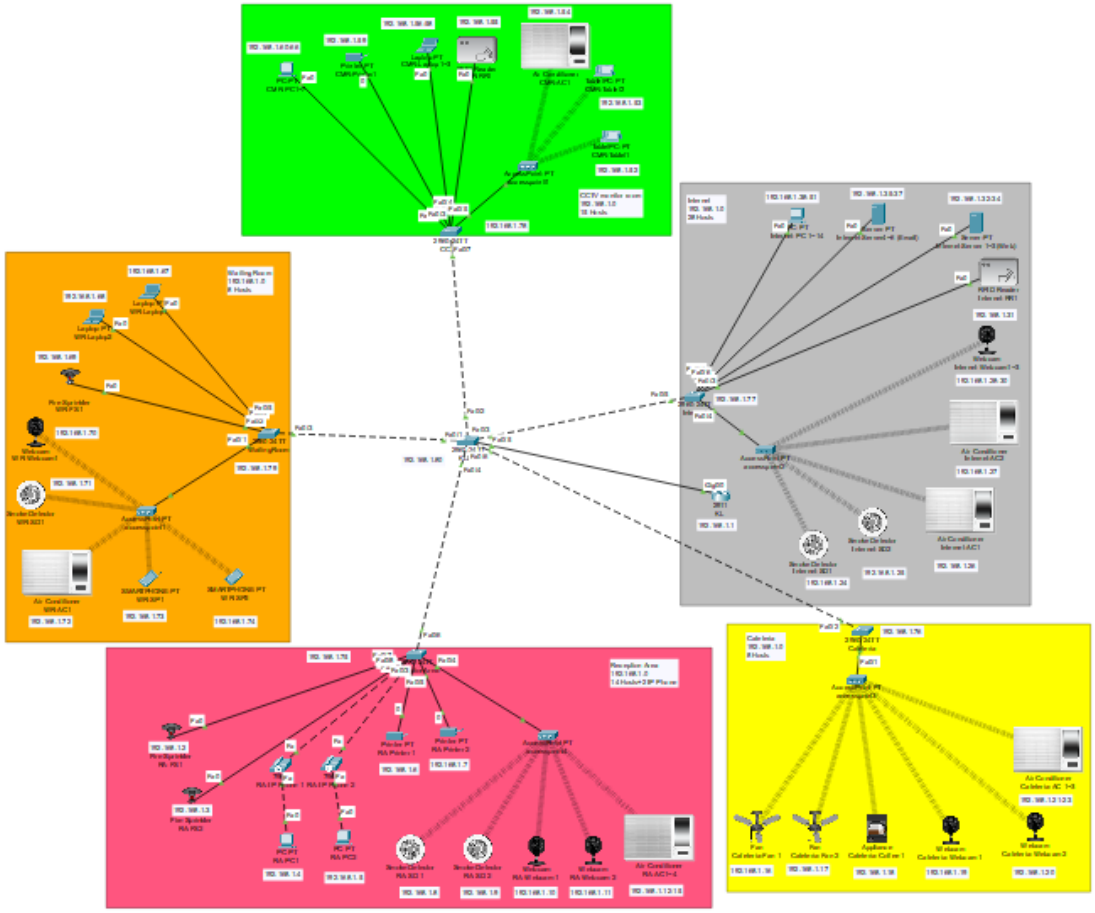


Figure 5.0 *Network Topology for KL Branch, First Floor*

The network topology employed on the first floor of the KL branch is an extended star topology. As shown in Figure 5.0, the network diagram of the first floor of KL building involves connecting multiple individual star networks by interconnecting their central hubs, allowing for increased coverage and flexibility in network design. The end devices of five departments are each connected to their respective switches, forming five small star topologies. Additionally, this arrangement promotes efficient resource allocation and minimizes the impact of potential network issues to specific departments, enhancing fault isolation. Furthermore, each switch is

then connected to a main switch named 'KLL1,' also known as the central hub, forming a larger star topology. This central hub configuration facilitates centralized network management, simplifying tasks such as software updates and security implementations across the entire floor. Moreover, the central hub will connect to a router named 'KL' to establish connections with other branches, ensuring seamless communication with Cyberjaya branch. In conclusion, the star topology is a suitable choice for our network, offering high reliability, optimal performance, centralized monitoring, ease of installation and expansion, and improved manageability.

5.2 IP addressing with justification

Subnetting is essential in networking for efficient IP address management, reduced broadcast traffic, and enhanced security. It optimizes network performance, simplifies administration, and supports scalability as networks expand. Subnetting is crucial for organizing and securing computer networks by isolating segments and facilitating effective IP address assignment.

Calculation:

Total Host : 73

$$2^n - 2 \geq 73$$

n=7

The total number of used hosts on the first floor of KL branch is 73. To determine the appropriate IP address for a network with 73 hosts, we need to consider the number of hosts and select a suitable subnet size. The closest power of 2 greater than 73 is 128 (2^7 = 128). This means the subnet size needs to accommodate at least 128 addresses.

32-7 = 25/prefix

11111111.11111111.11111111.10000000

255.255.255.128

Since the smallest subnet with more than 73 addresses is 128, it would need at least 7 bits for the host portion of the IP address (2^7 = 128). The remaining bits would be used for the network portion. Therefore, the subnet mask would be 255.255.255.128 (128 in binary is 10000000) with a prefix length of 25.

Assume that the first floor of KL branch is using 192.168.1.0 network:

Network Address		192.168.1.0
Subnet Mask		255.255.255.128
Usable IP Range	**First Valid Address (Default Gateway)**	192.168.1.1
	Second Valid Address	192.168.1.2
	Last Valid Address	192.168.1.126
Broadcast IP Address		192.168.1.127

Table 5.0 *IP Address Calculation*

This subnet allows for 126 usable IP addresses (128 total addresses minus the network and broadcast addresses), which is more than the total used hosts on the first floor of KL branch (73 hosts). 192.168.1.0 is used as network address because it is the first address in a subnet. Network address is used to represent the entire network. The last IP address 192.168.1.127 is designated as the broadcast address to send a message to all devices within the specific subnetwork. Also, the first valid IP address 192.168.1.1 is assigned to the router called ‘KL’, also known as default gateway.

Router	**IP Address**	**Subnet Mask**	**Default Gateway**
KL (Gigabit Ethernet 0/0)	192.168.1.1	255.255.255.128	
Switch	**IP Address**	**Subnet Mask**	**Default Gateway**
ReceptionArea	192.168.1.75	255.255.255.128	192.168.1.1
Cafeteria	192.168.1.76	255.255.255.128	192.168.1.1
Internet	192.168.1.77	255.255.255.128	192.168.1.1
CCTV	192.168.1.78	255.255.255.128	192.168.1.1
WaitingRoom	192.168.1.79	255.255.255.128	192.168.1.1
KLL1	192.168.1.80	255.255.255.128	192.168.1.1

Table 5.1 *IP Address of Router & Switch*

Hostname	Console Password	Password	Password (secret)	Banner
KLL1	vantech	kll1	kll12023	This is a secure system. Authorized Access Only!
Reception Area		receptionarea	receptionarea2023	
Cafeteria		cafeteria	cafeteria2023	
Internet		internet	internet2023	
CCTV		cctv	cctv2023	
Waiting Room		waitingroom	waitingroom2023	

Table 5.2 *Password for Switches and Router*

Department Name	Number of Hosts			Network IP	Subnet Mask	Range of Valid Hosts
	Used	Reserved	Total			
Reception area	14	11	25	192.168.1.0	255.255.255.128	192.168.1.2 - 192.168.1.15
Cafeteria	8	17	25			192.168.1.16- 192.168.1.23
Internet	28	7	35			192.168.1.24- 192.168.1.51
CCTV Monitor Room	15	5	20			192.168.1.52- 192.168.1.66
Waiting room	8	12	20			192.168.1.67- 192.168.1.74
Total	73	52	125			

Table 5.3 *Range and Total Number of Host*

Department	**Device Name**	**IP Address**	**Subnet Mask**	**Default Gateway**
Reception Area	RA-FS1	192.168.1.2	255.255.255.128	192.168.1.1
	RA-FS2	192.168.1.3	255.255.255.128	192.168.1.1
	RA-PC1	192.168.1.4	255.255.255.128	192.168.1.1
	RA-PC2	192.168.1.5	255.255.255.128	192.168.1.1
	RA-Printer 1	192.168.1.6	255.255.255.128	192.168.1.1
	RA-Printer 2	192.168.1.7	255.255.255.128	192.168.1.1
	RA-SD 1	192.168.1.8	255.255.255.128	192.168.1.1

	RA-SD2	192.168.1.9	255.255.255.128	192.168.1.1
	RA-Webcam 1	192.168.1.10	255.255.255.128	192.168.1.1
	RA-Webcam 2	192.168.1.11	255.255.255.128	192.168.1.1
	RA-AC1	192.168.1.12	255.255.255.128	192.168.1.1
	RA-AC2	192.168.1.13	255.255.255.128	192.168.1.1
	RA-AC3	192.168.1.14	255.255.255.128	192.168.1.1
	RA-AC4	192.168.1.15	255.255.255.128	192.168.1.1
	RA-IP Phone1			
	RA-IP Phone 2			
Cafeteria	Cafeteria-Fan1	192.168.1.16	255.255.255.128	192.168.1.1
	Cafeteria-Fan2	192.168.1.17	255.255.255.128	192.168.1.1
	Cafeteria-Coffee1	192.168.1.18	255.255.255.128	192.168.1.1
	Cafeteria-Webcam1	192.168.1.19	255.255.255.128	192.168.1.1
	Cafeteria-Webcam2	192.168.1.20	255.255.255.128	192.168.1.1
	Cafeteria-AC1	192.168.1.21	255.255.255.128	192.168.1.1
	Cafeteria-AC2	192.168.1.22	255.255.255.128	192.168.1.1
	Cafeteria-AC3	192.168.1.23	255.255.255.128	192.168.1.1
Internet	Internet-SD1	192.168.1.24	255.255.255.128	192.168.1.1
	Internet-SD1	192.168.1.25	255.255.255.128	192.168.1.1
	Internet-AC1	192.168.1.26	255.255.255.128	192.168.1.1
	Internet-AC2	192.168.1.27	255.255.255.128	192.168.1.1
	Internet-Webcam1	192.168.1.28	255.255.255.128	192.168.1.1
	Internet-Webcam2	192.168.1.29	255.255.255.128	192.168.1.1

	Internet-Webcam3	192.168.1.30	255.255.255.128	192.168.1.1
	Internet- RR1	192.168.1.31	255.255.255.128	192.168.1.1
	Internet-Server 1	192.168.1.32	255.255.255.128	192.168.1.1
	Internet-Server 2	192.168.1.33	255.255.255.128	192.168.1.1
	Internet-Server 3	192.168.1.34	255.255.255.128	192.168.1.1
	Internet-Server 4	192.168.1.35	255.255.255.128	192.168.1.1
	Internet-Server 5	192.168.1.36	255.255.255.128	192.168.1.1
	Internet-Server 6	192.168.1.37	255.255.255.128	192.168.1.1
	Internet-PC 1	192.168.1.38	255.255.255.128	192.168.1.1
	Internet-PC 2	192.168.1.39	255.255.255.128	192.168.1.1
	Internet-PC 3	192.168.1.40	255.255.255.128	192.168.1.1
	Internet-PC 4	192.168.1.41	255.255.255.128	192.168.1.1
	Internet-PC 5	192.168.1.42	255.255.255.128	192.168.1.1
	Internet-PC 6	192.168.1.43	255.255.255.128	192.168.1.1
	Internet-PC 7	192.168.1.44	255.255.255.128	192.168.1.1
	Internet-PC 8	192.168.1.45	255.255.255.128	192.168.1.1
	Internet-PC 9	192.168.1.46	255.255.255.128	192.168.1.1
	Internet-PC 10	192.168.1.47	255.255.255.128	192.168.1.1
	Internet-PC 11	192.168.1.48	255.255.255.128	192.168.1.1
	Internet-PC 12	192.168.1.49	255.255.255.128	192.168.1.1
	Internet-PC 13	192.168.1.50	255.255.255.128	192.168.1.1
	Internet-PC 14	192.168.1.51	255.255.255.128	192.168.1.1
CCTV Monitor Room	CMR-Tablet1	192.168.1.52	255.255.255.128	192.168.1.1
	CMR-Tablet2	192.168.1.53	255.255.255.128	192.168.1.1
	CMR-AC1	192.168.1.54	255.255.255.128	192.168.1.1

	CMR-RR1	192.168.1.55	255.255.255.128	192.168.1.1
	CMR-Laptop1	192.168.1.56	255.255.255.128	192.168.1.1
	CMR-Laptop2	192.168.1.57	255.255.255.128	192.168.1.1
	CMR-Laptop3	192.168.1.58	255.255.255.128	192.168.1.1
	CMR-Printer1	192.168.1.59	255.255.255.128	192.168.1.1
	CMR-PC1	192.168.1.60	255.255.255.128	192.168.1.1
	CMR-PC2	192.168.1.61	255.255.255.128	192.168.1.1
	CMR-PC3	192.168.1.62	255.255.255.128	192.168.1.1
	CMR-PC4	192.168.1.63	255.255.255.128	192.168.1.1
	CMR-PC5	192.168.1.64	255.255.255.128	192.168.1.1
	CMR-PC6	192.168.1.65	255.255.255.128	192.168.1.1
	CMR-PC7	192.168.1.66	255.255.255.128	192.168.1.1
Waiting Room	WR-Laptop1	192.168.1.67	255.255.255.128	192.168.1.1
	WR-Laptop2	192.168.1.68	255.255.255.128	192.168.1.1
	WR-FS1	192.168.1.69	255.255.255.128	192.168.1.1
	WR-Webcam1	192.168.1.70	255.255.255.128	192.168.1.1
	WR-SD1	192.168.1.71	255.255.255.128	192.168.1.1
	WR-AC1	192.168.1.72	255.255.255.128	192.168.1.1
	WR-SP1	192.168.1.73	255.255.255.128	192.168.1.1
	WR-SP2	192.168.1.74	255.255.255.128	192.168.1.1

Table 5.4 *IP Address of Devices (expect Switches and Router)*

6.0 Kuala Lumpur Second Floor

Lee Yan Sheng (21053301)

6.1 Topology with justification

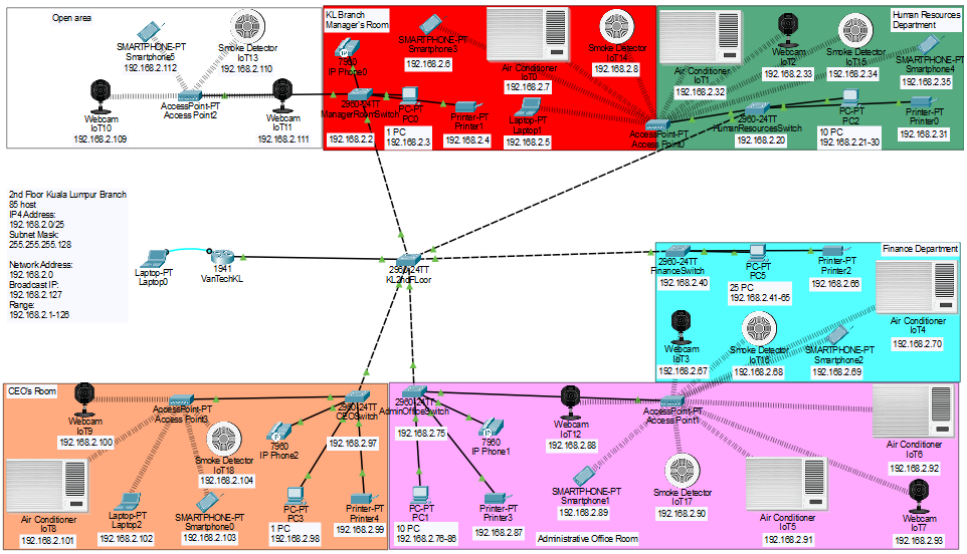


Figure 6.0 *Network Topology for KL Branch, Second Floor*

There are two types of network topologies: logical and physical. Physical topologies display the network's precise connection patterns and physical wired layout. The logical network path that data takes to get from one end to the other is displayed by logical topologies. Bus, ring, star, and mesh topologies are a few of the most common network topologies.

The topology system that is most frequently used is star topology. In this configuration, every node is connected to a hub, switch, or computer, which is the central network device. Because star topology is centralized, it is dependable, easy to use, and manage. Star topology, however, is expensive and requires ongoing upkeep.

6.2 IP addressing with justification

At VanTech's Kuala Lumpur branch, the second floor accommodates approximately 85 hosts, necessitating a meticulous calculation of IP addresses and subnet masks to efficiently manage and optimize the network infrastructure. This task involves devising a robust addressing scheme that not only accommodates the current number of hosts but also allows for scalability and efficient utilization of available IP address space. The implementation of an effective IP addressing strategy is essential for fostering a resilient and well-organized networking environment, ensuring seamless communication, and facilitating future expansion and technological advancements within the branch. Therefore, calculations are needed.

Calculation:

$2^n-2 >= 85$

n=7

32-7 =25

Binary form:

11111111.11111111.11111111.10000000

This is a Class C network

Network Address		192.168.2.0
Subnet Mask		255.255.255.128
Usable IP Range	**First Valid Address (Default Gateway)**	192.168.2.1
	Second Valid Address	192.168.2.2
	Last Valid Address	192.168.2.126
Broadcast IP Address		192.168.2.127

Table 6.0 *IP Address Calculation*

Router and Switch Device	IP Address	Subnet Mask	Default Gateway
VanTechKL (Router) (Gigabit Ethernet 0/1)	192.168.2.1	255.255.255.128	-
KL Branch Manager's Room (Switch)	192.168.2.2	255.255.255.128	192.168.2.1
Human Resources (Switch)	192.168.2.20	255.255.255.128	192.168.2.1
Finance (Switch)	192.168.2.40	255.255.255.128	192.168.2.1
Administrative Office (Switch)	192.168.2.75	255.255.255.128	192.168.2.1
CEO (Switch)	192.168.2.97	255.255.255.128	192.168.2.1

Table 6.1 *IP Address of Router & Switch*

Location	Device	IP Address	Subnet Mask	Default Gateway
KL Branch Manager's Room	PC	192.168.2.3	255.255.255.128	192.168.2.1
	Printer	192.168.2.4	255.255.255.128	192.168.2.1
	Laptop	192.168.2.5	255.255.255.128	192.168.2.1
	Smartphone	192.168.2.6	255.255.255.128	192.168.2.1
	Air Conditioner	192.168.2.7	255.255.255.128	192.168.2.1
	Smoke Detector	192.168.2.8	255.255.255.128	192.168.2.1
Human Resources Department	10 PCs	192.168.2.21-30	255.255.255.128	192.168.2.1
	Printer	192.168.2.31	255.255.255.128	192.168.2.1
	Air Conditioner	192.168.2.32	255.255.255.128	192.168.2.1
	Webcam	192.168.2.33	255.255.255.128	192.168.2.1
	Smoke Detector	192.168.2.34	255.255.255.128	192.168.2.1
	Smartphone	192.168.2.35	255.255.255.128	192.168.2.1
Finance Department	25 PCs	192.168.2.41-65	255.255.255.128	192.168.2.1
	Printer	192.168.2.66	255.255.255.128	192.168.2.1
	Webcam	192.168.2.67	255.255.255.128	192.168.2.1
	Smoke Detector	192.168.2.68	255.255.255.128	192.168.2.1
	Smartphone	192.168.2.69	255.255.255.128	192.168.2.1
	Air Conditioner	192.168.2.70	255.255.255.128	192.168.2.1
Administrative Office Room	10 PCs	192.168.2.76-86	255.255.255.128	192.168.2.1
	Printer	192.168.2.87	255.255.255.128	192.168.2.1
	Webcam	192.168.2.88	255.255.255.128	192.168.2.1
	Smartphone	192.168.2.89	255.255.255.128	192.168.2.1
	Smoke Detector	192.168.2.90	255.255.255.128	192.168.2.1

	Air Conditioner	192.168.2.91	255.255.255.128	192.168.2.1
	Air Conditioner 2	192.168.2.92	255.255.255.128	192.168.2.1
	Webcam 2	192.168.2.93	255.255.255.128	192.168.2.1
CEO's Room	PC	192.168.2.98	255.255.255.128	192.168.2.1
	Printer	192.168.2.99	255.255.255.128	192.168.2.1
	Webcam	192.168.2.100	255.255.255.128	192.168.2.1
	Air Conditioner	192.168.2.101	255.255.255.128	192.168.2.1
	Laptop	192.168.2.102	255.255.255.128	192.168.2.1
	Smartphone	192.168.2.103	255.255.255.128	192.168.2.1
	Smoke Detector	192.168.2.104	255.255.255.128	192.168.2.1
Open Area	Webcam	192.168.2.109	255.255.255.128	192.168.2.1
	Smoke Detector	192.168.2.110	255.255.255.128	192.168.2.1
	Webcam 2	192.168.2.111	255.255.255.128	192.168.2.1
	Smartphone	192.168.2.112	255.255.255.128	192.168.2.1

Table 6.2 *IP Address of Devices (expect Switches and Router)*

Name	Device	Secret	Password	Banner
KL 2nd Floor	Switch	KLBranch123	KLBranch123@	This is VanTech Kuala Lumpur 2nd Floor. Unauthorized Access is denied.
Manager Room Switch	Switch	VanTechKL	VanTechKL	-
Human Resources Switch	Switch	VanTechKL	VanTechKL	-
Finance Switch	Switch	VanTechKL	VanTechKL	-
Admin Office Switch	Switch	VanTechKL	VanTechKL	-
CEO Switch	Switch	VanTechKL	VanTechKL	-

Table 6.3 *Password for Switches and Router*

7.0 Cyberjaya First Floor

Ooi Jin Yon 22039069

7.1 Topology with justification

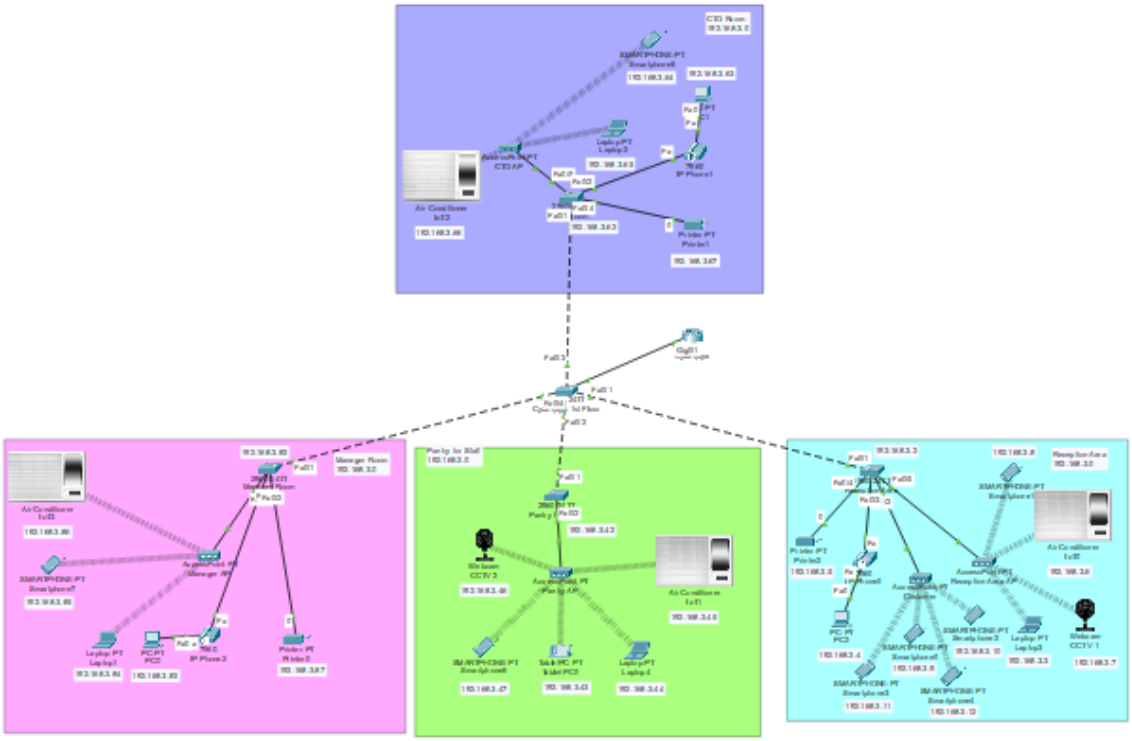


Figure 7.0 *Network Topology for Cyberjaya Branch, First Floor*

We used extended star topology to connect all end devices on the first floor of the Cyberjaya branch. The reason we chose this topology is because this is the most common type of network topology which is used in different sectors. It is a type of network topology in which all the nodes are connected to one switch and then that switch is directly connected to a central switch (Gupta, 2023).

Based on figure 7.0, this is clearly show that there is a main switch which connected to the router and other switches from the reception area, pantry, CTO room and Manager room. In each section, they also connected to different end devices. Although they are in different rooms, they can still communicate via the room switch.

7.2 IP addressing with justification

Calculation	$2^n \geq 100$ $n = 7$ $2^7 = 128$ 32-7=25
Binary Form	11111111 11111111 11111111 10000000 (/25)
Subnet Mask	255.255.255.128
Usable Host	$2^7 - 2 = 126$

Table 7.0 *IP Address Calculation*

Network Address		192.168.3.0
Subnet Mask		255.255.255.128
Usable IP Range	**First Valid Address (Default Gateway)**	192.168.3.1
	Second Valid Address	192.168.3.2
	Last Valid Address	192.168.3.126
Broadcast IP Address		192.168.3.127

Table 7.1 *Subnet Table*

Department Name	Hosts			Network IP	Subnet Mask	Range of Valid Hosts
	Used	Reserve	Total			
Reception area	11	29	40	192.168.3.0	255.255.255.128	192.168.3.2 - 192.168.3.41
Pantry for Staff	6	14	20			192.168.3.42 - 192.168.3.61
CTO Room	6	14	20			192.168.3.62 - 192.168.3.81
Manager Room	6	14	20			192.168.3.82- 192.168.3.101

Table 7.2 *Range and Total Number of Host*

Router and Switch Device	IP Address	Subnet Mask	Default Gateway
Cyberjaya (Router) (Gigabit Ethernet 0/1)	192.168.3.1	255.255.255.128	-
Reception Area (Switch)	192.168.3.2	255.255.255.128	192.168.3.1
Pantry for Staff (Switch)	192.168.3.42	255.255.255.128	192.168.3.1
CTO Room (Switch)	192.168.3.62	255.255.255.128	192.168.3.1
Manager Room (Switch)	192.168.3.82	255.255.255.128	192.168.3.1

Table 7.3 *IP Address of Router & Switch*

Location	Device Name	IP Address	Subnet Mask	Default Gateway
Reception Area	Laptop3	192.168.3.3	255.255.255.128	192.168.3.1
	PC2(1)	192.168.3.4	255.255.255.128	192.168.3.1
	Printer2(1)	192.168.3.5	255.255.255.128	192.168.3.1
	IoT0(1)	192.168.3.6	255.255.255.128	192.168.3.1
	CCTV 1	192.168.3.7	255.255.255.128	192.168.3.1
	Smartphone1(1)	192.168.3.8	255.255.255.128	192.168.3.1
	Smartphone0(1)	192.168.3.9	255.255.255.128	192.168.3.1
	Smartphone2(1)	192.168.3.10	255.255.255.128	192.168.3.1
	Smartphone3(1)	192.168.3.11	255.255.255.128	192.168.3.1
	Smartphone4(1)	192.168.3.12	255.255.255.128	192.168.3.1
Pantry for Staff	Tablet PC0	192.168.3.43	255.255.255.128	192.168.3.1
	Laptop4	192.168.3.44	255.255.255.128	192.168.3.1
	IoT1(1)	192.168.3.45	255.255.255.128	192.168.3.1
	CCTV 2	192.168.3.46	255.255.255.128	192.168.3.1
	Smartphone6	192.168.3.47	255.255.255.128	192.168.3.1
CTO Room	PC1(1)	192.168.3.63	255.255.255.128	192.168.3.1
	Smartphone5(1)	192.168.3.64	255.255.255.128	192.168.3.1
	Laptop2(1)	192.168.3.65	255.255.255.128	192.168.3.1
	IoT2(1)	192.168.3.66	255.255.255.128	192.168.3.1
	Printer1(1)	192.168.3.67	255.255.255.128	192.168.3.1
Manager Room	PC0(1)	192.168.3.83	255.255.255.128	192.168.3.1
	Laptop1(1)	192.168.3.84	255.255.255.128	192.168.3.1
	Smartphone7	192.168.3.85	255.255.255.128	192.168.3.1

	IoT3(1)	192.168.3.86	255.255.255.128	192.168.3.1
	Printer0(1)	192.168.3.87	255.255.255.128	192.168.3.1

Table 7.4 *IP Address of Devices (expect Switches and Router)*

Name	**Device**	**Secret**	**Password**	**Banner**
Cyberjaya First Floor	Switch	Cyberjaya123	Cyberjaya	This is VanTech Cyberjaya Branch Ground Floor
Reception Area	Switch	-	Cyberjaya2023	-
Pantry	Switch	-	Cyberjaya2023	-
CTO	Switch	VanTech003	VanTech003	-
Manager	Switch	VanTech003	VanTech003	-

Table 7.5 *Password for Switches and Router*

On the Cyberjaya branch ground floor, we estimated there will be at least 100 hosts in the coming year. In this case, we decided to give more reserved hosts for future use. The total number of used hosts on the first floor of this branch is 29, and the reserve host is 71. To determine the appropriate IP address for a network with 100 hosts, we need to consider the number of hosts and select a suitable subnet size. Table 7.2 shows the calculation of the subnet mask, of which 128 is most suitable for this floor. Besides that, table 7.3 shows the IP address range that we will use for Cyberjaya on the first floor. After that, we separate a range for each location or room so we can easily track the IP address (Table 7.4). This help to

All switches are connected to router in Cyberjaya. As there have different default gateway in each location, but their switches default gateway is the same, which is 192.168.3.1. In

addition, the other switches IP address was also show on Table 7.5. In short, all devices can connect router via different switches. The table 7.6 show all device IP address for Cyberjaya 1st floor with their default gateway. At the same time, table 7.7 show the password, secret and banner use for this floor which provides a more secure and safe guarantee for the company.

8.0 Cyberjaya Second Floor

Tan Ze Hao 22014690

8.1 Topology with justification

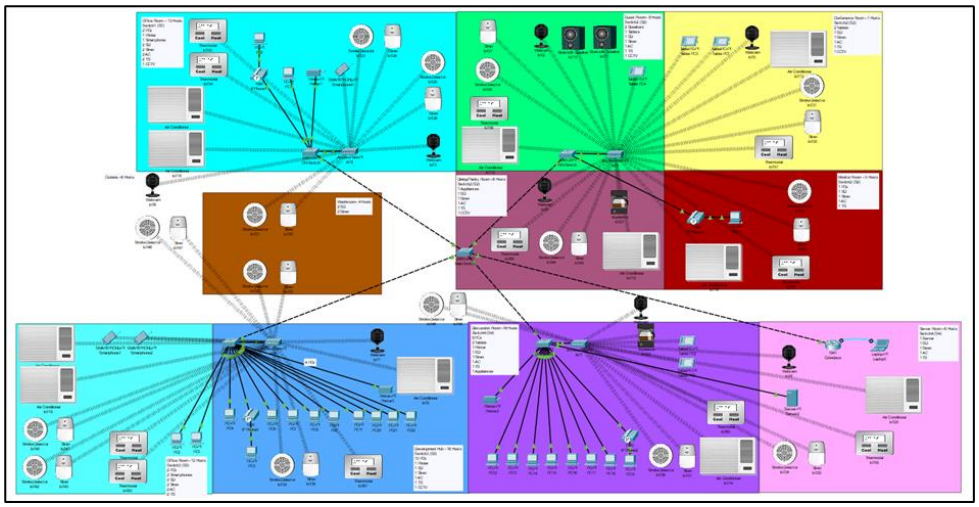


Figure 8.0: *Network Topology for Cyberjaya Branch, Second Floor*

The image above shows the network topology for Cyberjaya branch, second floor. Overall, the floor consists of a total of 23 PCs, 15 smoke detectors, 15 sirens, 11 air conditioners, 11 thermostats, 8 CCTVs, 5 tablets, 3 printers, 3 smartphones, and 2 coffee machines. All the devices are connected to their respective switches to minimize cable lengths (Vertix, 2021). Each individual room is placed with an air conditioner to maintain office temperature so that the office equipment can work at maximum efficiency (Synecore, 2019). All the devices are connected to the network with either cables or via wireless connections.

8.2 IP Addressing with justification

Total Hosts: 99

2^n – 2 >= 99

n = 7

32 -7 = 25

11111111.11111111.11111111.10000000

= 255.255.255.128 (Subnet Mask)

2^7 – 2 = 126 usable IP addresses

Subnet Mask	255.255.255.128
Network Address	192.168.4.0
First valid IP address (Default Gateway)	192.168.4.1
Last valid IP address	192.168.4.126
Broadcast address	192.168.4.127

Table 8.0: *IP Address Calculation*

Table 8.0 shows the available IP that can be assigned to Cyberjaya branch, second floor. With the first valid IP being 192.168.4.1 and last valid IP being 192.168.4.126. Additional information such as the subnet mask, network address and broadcast address are also displayed in table 8.0.

Device	IP Address	Subnet Mask	Default Gateway
Cyberjaya-Router	192.168.4.1	255.255.255.128	-
DH-Switch	192.168.4.3		192.168.4.1
DR-Switch	192.168.4.20		
GR-Switch	192.168.4.27		
OR-Switch	192.168.4.40		

Table 8.1: *IP Address of Router & Switch*

The table above shows all the IP addresses used by each switch and router on the second floor of Cyberjaya branch. Static Ips are used to assigned for all switches and router. There are a total of four switches and one router with the same subnet mask of 255.255.255.128. All the switches are also assigned with a default gateway which links back to the router.

Devices	Password	Secret	Banner MOTD
Cyberjaya-Router	Cyberfloor2	Cyberfloor2nd	Authorized personnel only
DH-Switch	Cyberfloor2	Cyberfloor2nd	Authorized personnel only
DR-Switch	Cyberfloor2	Cyberfloor2nd	Authorized personnel only
GR-Switch	Cyberfloor2	Cyberfloor2nd	Authorized personnel only
OR-Switch	Cyberfloor2	Cyberfloor2nd	Authorized personnel only

Table 8.2: *Password for Switches and Router*

Table 8.2 shows a list of the router and all the switches that are set with passwords for an extra layer of security and encryption purposes. There will be a message displayed warning that it is for authorized personnel only when trying to access the configuration of the router and switches.

Location	Device	IP Address	Subnet Mask	Default Gateway
Development Hub	10 PCs	192.168.4.4 - 192.168.4.13	255.255.255.128	192.168.4.1
	1 Printer	192.168.4.14		
	1 Smoke Detector	192.168.4.15		
	1 Siren	192.168.4.16		
	1 Thermostat	192.168.4.17		
	1 Air Conditioner	192.168.4.18		
	1 Webcam	192.168.4.19		
Server Room	1 Server	192.168.4.21	255.255.255.128	
	1 Smoke Detector	192.168.4.22		
	1 Siren	192.168.4.23		
	1 Thermostat	192.168.4.24		
	1 Air Conditioner	192.168.4.25		
	1 Webcam	192.168.4.26		
Monitor Room	1 PC	192.168.4.28	255.255.255.128	
	1 Smoke Detector	192.168.4.29		
	1 Siren	192.168.4.30		
	1 Thermostat	192.168.4.31		
	1 Air Conditioner	192.168.4.32		
Conference Room	2 Tablets	192.168.4.33 - 192.168.4.34	255.255.255.128	
	1 Webcam	192.168.4.35	255.255.255.128	
	1 Air	192.168.4.36		

	Conditioner			
	1 Smoke Detector	192.168.4.37		
	1 Siren	192.168.4.38		
	1 Thermostat	192.168.4.39		
Office Room 1	2 PCs	192.168.4.41 - 192.168.4.42	255.255.255.128	
	1 Printer	192.168.4.43		
	1 Smartphone	192.168.4.44		
	2 Air Conditioner	192.168.4.45 - 192.168.4.46		
	2 Thermostat	192.168.4.47 - 192.168.4.48		
	2 Smoke Detectors	192.168.4.49 & 192.168.4.51		
	2 Siren	192.168.4.50 & 192.168.4.52		
	1 Webcam	192.168.4.53		
Office Room 2	2 PCs	192.168.4.54 - 192.168.4.55	255.255.255.128	
	2 Thermostat	192.168.4.56 - 192.168.4.57		
	2 Siren	192.168.4.58 - 192.168.4.59		
	2 Smoke Detectors	192.168.4.60 - 192.168.4.61		
	2 Air Conditioners	192.168.4.62 - 192.168.4.63		
	2 Smartphones	192.168.4.64 - 192.168.4.65		
Guest Room	1 Air Conditioner	192.168.4.66	255.255.255.128	
	1 Thermostat	192.168.4.67		
	1 Smoke Detector	192.168.4.68		
	1 Siren	192.168.4.69		
	1 Webcam	192.168.4.70		
	2 Speaker	192.168.4.71 -		

		192.168.4.72		
	1 Tablet	192.168.4.73		
Dining/Pantry Room	1 Coffee Machine	192.168.4.74	255.255.255.128	
	1 Air Conditioner	192.168.4.75		
	1 Siren	192.168.4.76		
	1 Smoke Detector	192.168.4.77		
	1 Thermostat	192.168.4.78		
	1 Webcam	192.168.4.79		
Discussion Room	1 Printer	192.168.4.80	255.255.255.128	
	8 PCs	192.168.4.81 - 192.168.4.88		
	1 Smoke Detector	192.168.4.89		
	1 Siren	192.168.4.90		
	1 Air Conditioner	192.168.4.91		
	1 Thermostat	192.168.4.92		
	2 Tablets	192.168.4.93 - 192.168.4.94		
	1 Coffee Machine	192.168.4.95		
Washroom	2 Smoke Detector	192.168.4.96 & 192.168.4.98	255.255.255.128	
	2 Siren	192.168.4.97 & 192.168.4.99		
Outside	2 Webcam	192.168.4.100 - 192.168.4.101	255.255.255.128	
	2 Smoke Detector	192.168.4.102 & 192.168.4.104		
	2 Siren	192.168.4.103 & 192.168.4.105		

Table 8.3: *IP Address of Devices (expect Switches and Router)*

Table 8.3 shows all the IP addresses assigned to each room on the second floor of Cyberjaya branch. The total number of hosts is 99. Static IP is used to assign IP addresses for all hosts. Starting from 192.168.4.4 assigned in the development hub and ending with 192.168.4.105 assigned at the hallway outside of all the rooms. The remaining number of hosts reserved for the second floor of Cyberjaya branch is 21. The subnet mask and default gateway of all devices is set to 255.255.255.128 and 192.168.4.1 respectively.

9.0 Configuration

9.1 Router Configuration & Connection

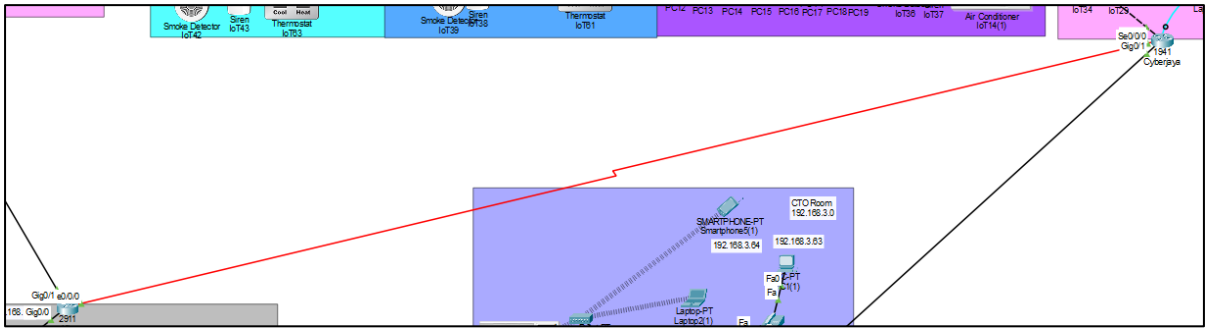
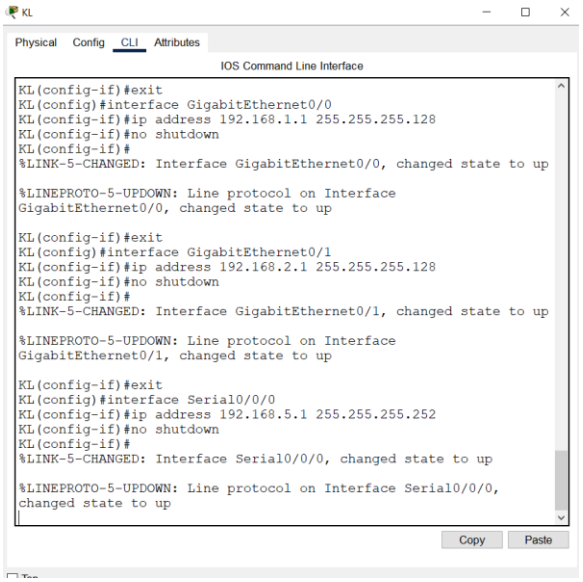


Figure 9.0 *Router Connection*

Each building has a router in the server room connecting two floors, and the routers in KL and Cyberjaya are linked through a serial cable. A serial cable is commonly used to provide direct and reliable connectivity between routers for configuration, troubleshooting, and out-of-band management purposes.

Router	Interface	IP Address	Subnet Mask
KL	S0/0/0	192.168.5.1	255.255.255.252
	G0/0	192.168.1.1	255.255.255.128
	G0/1	192.168.2.1	
Cyberjaya	S0/0/0	192.168.5.2	255.255.255.252
	G0/0	192.168.4.1	255.255.255.128
	G0/1	192.168.3.1	

Table 9.0 *IP Addressing Table*



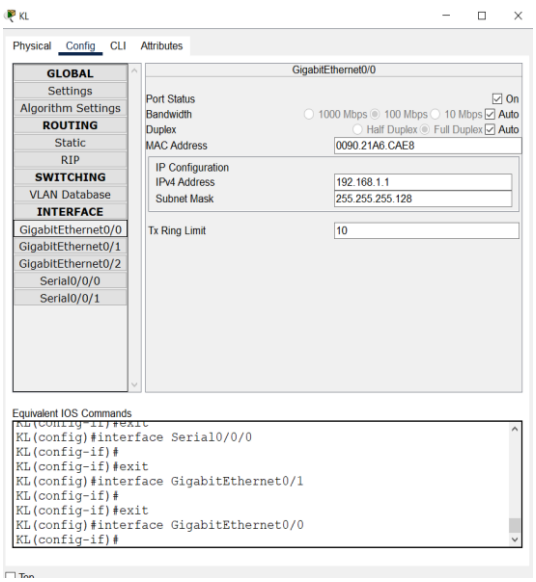
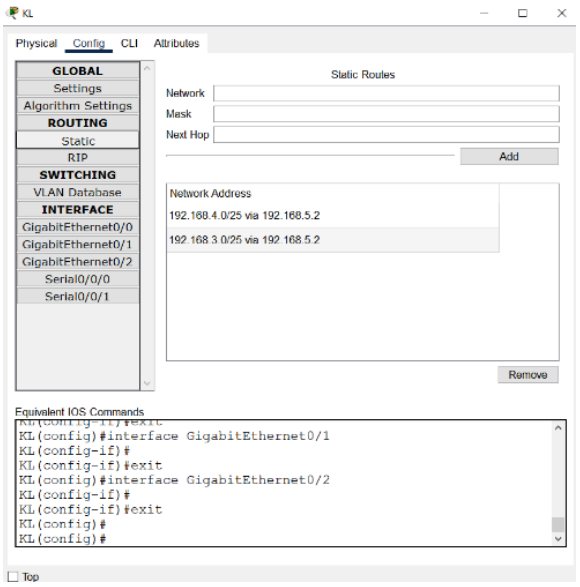


Figure 9.1 *IP Address Configuration (CLI)* ***Figure 9.2*** *IP Address Configuration (Config)*

As shown in the figures above, the KL router has three interfaces, respectively connecting to the 1st floor in KL (192.168.1.1), the 2nd floor in KL (192.168.2.1), and the router in Cyberjaya (192.168.5.1).

For example, as shown in the Figure 9.1 , navigate to the router's CLI tab and enter the following commands: "interface GigabitEthernet0/1" "IP address 192.168.2.1 255.255.255.128" "no shutdown" to assign Ip address to the GigabitEthernet0/1 interface.



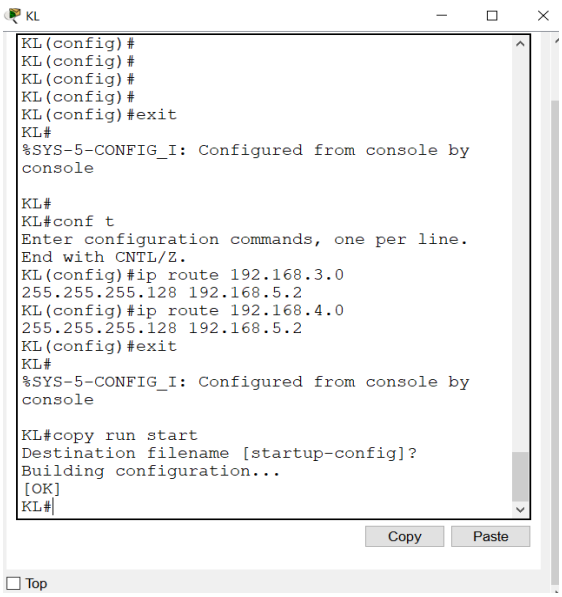


Figure 9.3 *Routing (KL Router Config)*

Figure 9.4 *Routing (KL Router CLI)*

Routing is the process of selecting a path to deliver data to its destination. Any host capable of generating a routing table can perform routing, and this process becomes necessary when delivering data packets to a network that is not directly connected to the sender.

For the KL router, the networks that are not directly connected are on the first and second floors of the Cyberjaya branch. As the KL router is connected to the Cyberjaya router via the s0/0/0 interface, the next hop is 192.168.5.2. Therefore, in the router's CLI tab, enter the following commands:

- "IP route 192.168.3.0 255.255.255.128 192.168.5.2"
- "IP route 192.168.4.0 255.255.255.128 192.168.5.2"

9.2 Ping

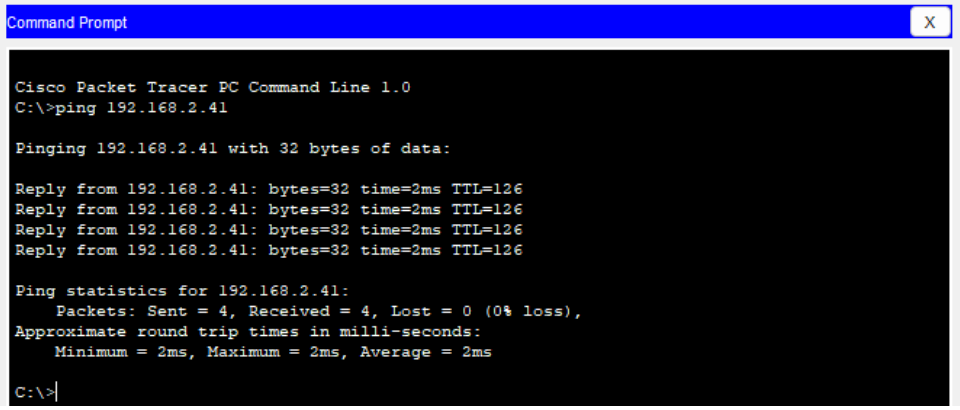


Figure 9.5 *Ping from Cyberjaya 2nd floor to Kuala Lumpur 2nd floor*

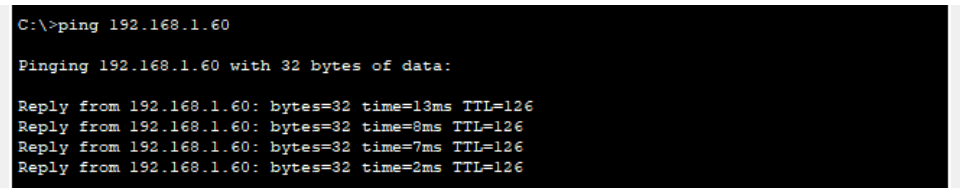


Figure 9.6 *Ping from Cyberjaya 1st floor to Kuala Lumpur 1st floor*

Figure 9.5 and figure 9.6 show that all the devices are connected to the network through their corresponding wires and wireless connections. Both the ping test from figure 9.5 and figure 9.6 illustrate that the devices from Cyberjaya 1st floor and Cyberjaya 2nd floor have gotten successful pinging replies from the devices in Kuala Lumpur 1st floor and Kuala Lumpur 2nd floor.

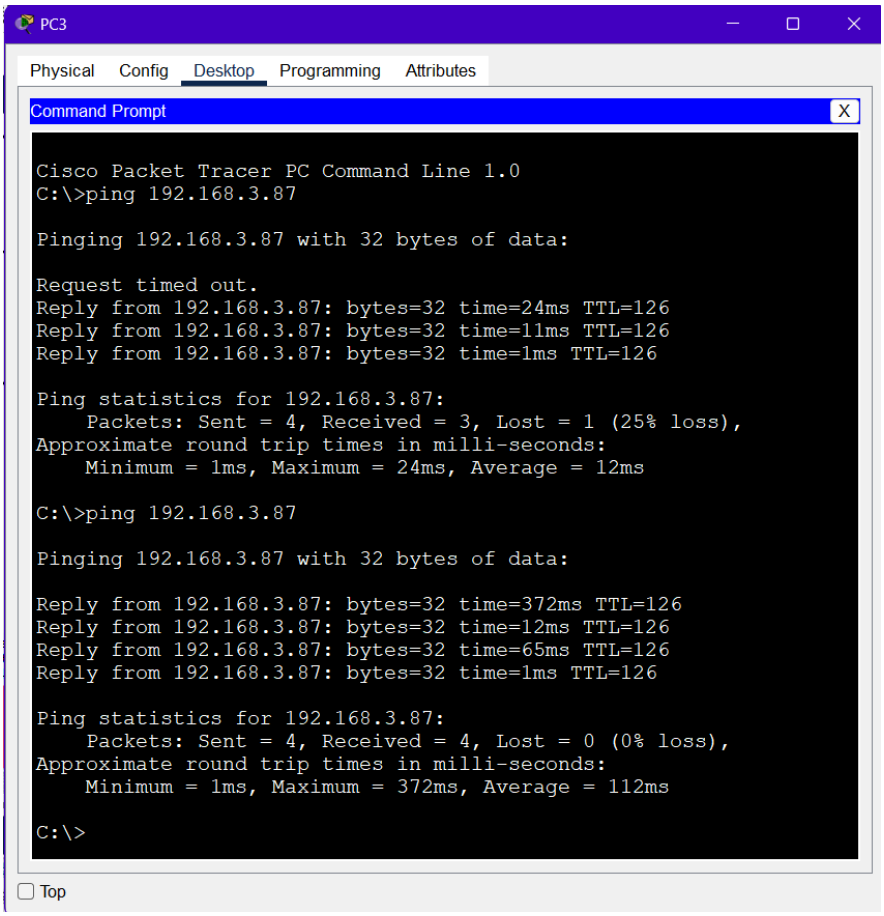


Figure 9.7 *Ping from Kuala Lumpur 2nd Floor to Cyberjaya 1st Floor*

Figure 9.7 shows subsequently that the PCs from Kuala Lumpur's 2nd Floor can ping Cyberjaya's 1st Floor PCs, thus receiving replies from the pinged IP. However, due to the simulation error, there appears to be a 25% packet loss the first time it is pinged.

9.3 Server (Email, Web, DNS)

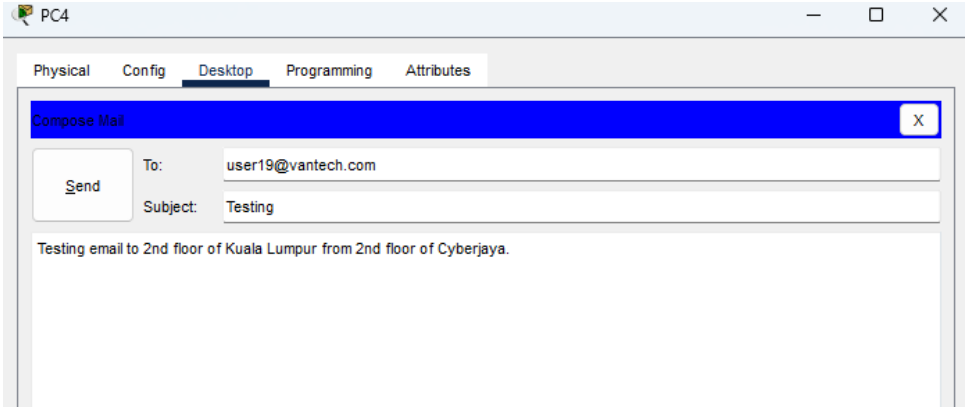


Figure 9.8 *Email from Cyberjaya 2nd floor to Kuala Lumpur 2nd floor*

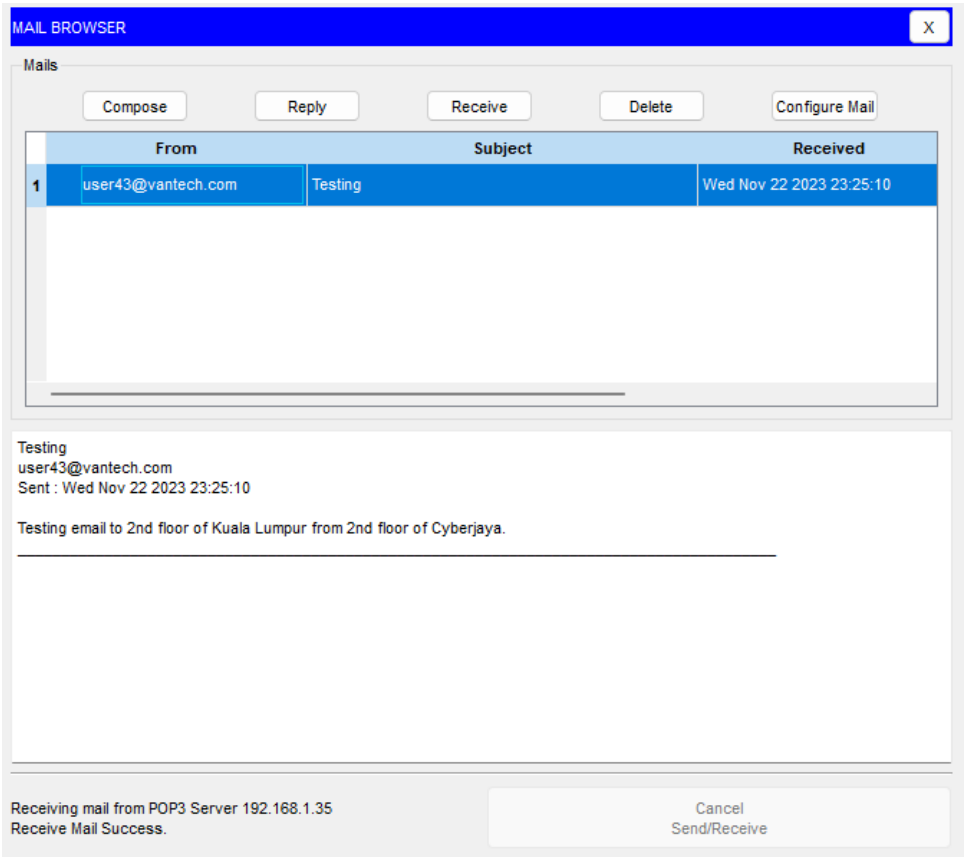


Figure 9.9 *Email received from Cyberjaya 2nd floor at Kuala Lumpur 2nd floor*

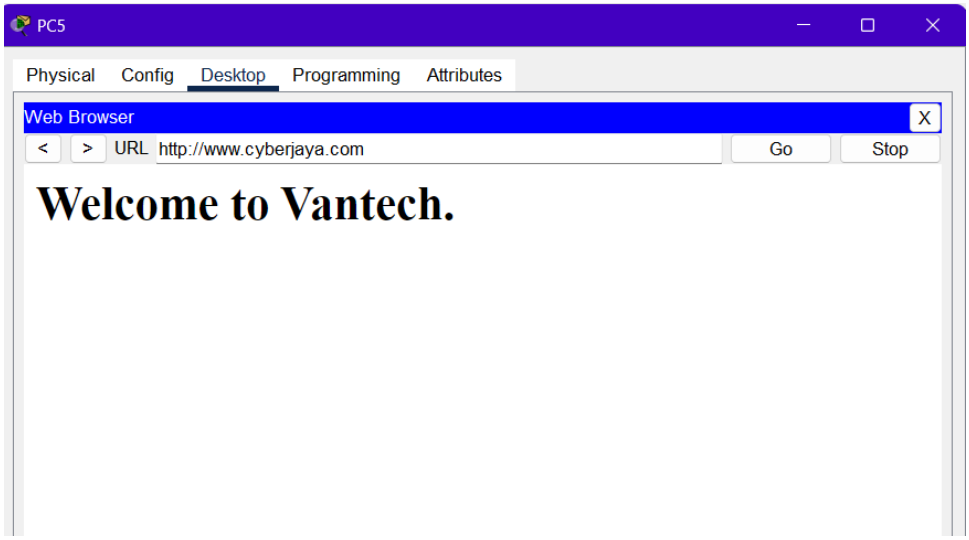


Figure 9.10 *Accessing the Cyberjaya website.*

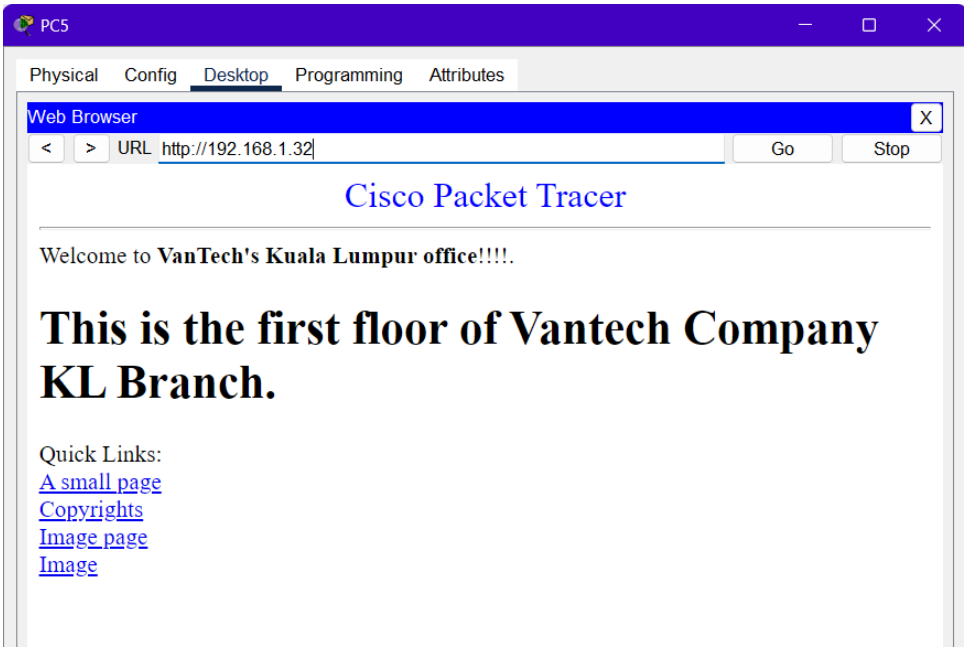


Figure 9.11 *Accessing the Kuala Lumpur website.*

10.0 Conclusion

In conclusion, we as a technical team involved successful build a network design for VanTech's Kuala Lumpur and Cyberjaya branches with the requirements and objectives. The combination of extended star topology provides scalable and easily manageable infrastructure. Besides that, the layout for each building is taken into consideration and ensure connectivity between all departments in different branches.

From this experience, we have learned about different type of topology, where the star topology is more reliable and efficient. Besides that, we know to calculate the subnet mask, network address and the valid host by using the formula **"$2^n - 2 >$ Require host"**. Moreover, we learned to connect 4 different floor using router and switches, where require to configurate with different step by assigning IP addresses and giving them password to protect company's data.

This assignment involved a thorough exploration of networking concepts, covering IP address configuration, switch and router setup, network diagram design, subnet mask calculations, and floorplan design. Personally designing the floor plan and network diagram, along with applying configuration methods learned in class, deepened my understanding of networking. The assignment highlighted the importance of using appropriate IP addresses and subnet masks, emphasizing the significance of networking within the structure. Overall, the assignment offered a comprehensive approach, integrating theoretical knowledge with practical skills.

11.0 References

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12.0 Peer Evaluation

PEER EVALUATION FORM

***To be filled up by individual student and include with soft copy the document.*

Student Name / ID Number : Lim Jia Xuan /2211078

Please answer 'Yes' / 'No'

	Myself ID NO: 22110787	**Member 2 ID NO: 21053301**	**Member 3 ID NO: 22039069**	**Member 4 ID NO: 22014690**
Participated in group discussion	Yes	Yes	Yes	Yes
Helped to monitor group progress	Yes	Yes	Yes	Yes
Contributed useful ideas	Yes	Yes	Yes	Yes
Completed work on time	Yes	Yes	Yes	Yes
Submitted good quality of work	Yes	Yes	Yes	Yes
Communicated effectively with group members	Yes	Yes	Yes	Yes
Helped others with their work when needed	Yes	Yes	Yes	Yes

Good relationship with group members	Yes	Yes	Yes	Yes
Overall was a valuable member of the team	Yes	Yes	Yes	Yes

Self-Reflection *(what I did best and what I should improve in future):*

In this assignment, I believe I have actively engaged in continuous discussions with team members, providing ideas, and participating in every consultation session to improve our assignment. Similarly, I have sought to meet the team's requirements as much as possible and contributed to advancing the progress of the task. I acknowledge the areas for improvement, particularly in enhancing my knowledge of subnetting and IPv6, to enhance the overall quality of the assignment.

PEER EVALUATION FORM

***To be filled up by individual student and include with soft copy the document.*

Student Name / ID Number : Lee Yan Sheng / 21053301

Please answer 'Yes' / 'No'

	Myself ID NO: 21053301	**Member 2 ID NO: 22110787**	**Member 3 ID NO: 22039069**	**Member 4 ID NO: 22014690**
Participated in group discussion	Yes	Yes	Yes	Yes
Helped to monitor group progress	Yes	Yes	Yes	Yes
Contributed useful ideas	Yes	Yes	Yes	Yes
Completed work on time	Yes	Yes	Yes	Yes
Submitted good quality of work	Yes	Yes	Yes	Yes
Communicated effectively with group members	Yes	Yes	Yes	Yes
Helped others with their work when needed	Yes	Yes	Yes	Yes

Good relationship with group members	Yes	Yes	Yes	Yes
Overall was a valuable member of the team	Yes	Yes	Yes	Yes

Self-Reflection *(what I did best and what I should improve in future):*

I provided ideas and feedback to contribute to the team. However, I found that I should improve my overall network planning skills in the future.

PEER EVALUATION FORM

***To be filled up by individual student and include with soft copy the document.*

Student Name / ID Number : Ooi Jin Yon / 22039069

Please answer 'Yes' / 'No'

	Myself ID NO: 22039069	**Member 2 ID NO: 22110787**	**Member 3 ID NO: 21053301**	**Member 4 ID NO: 22014690**
Participated in group discussion	Yes	Yes	Yes	Yes
Helped to monitor group progress	Yes	Yes	Yes	Yes
Contributed useful ideas	Yes	Yes	Yes	Yes
Completed work on time	Yes	Yes	Yes	Yes
Submitted good quality of work	Yes	Yes	Yes	Yes
Communicated effectively with group members	Yes	Yes	Yes	Yes
Helped others with their work when needed	Yes	Yes	Yes	Yes

Good relationship with group members	Yes	Yes	Yes	Yes
Overall was a valuable member of the team	Yes	Yes	Yes	Yes

Self-Reflection *(what I did best and what I should improve in future):*

First of all, I think our team have done our best on our own part and have more discussion during this period. I should improve in learning more end device setup like TV, and IoT. Besides that, I can learn more about IPv6 and server setup where they are not included in my part.

PEER EVALUATION FORM

***To be filled up by individual student and include with soft copy the document.*

Student Name / ID Number : Tan Ze Hao / 22014690

Please answer 'Yes' / 'No'

	Myself ID NO: 22110787	**Member 2 ID NO: 21053301**	**Member 3 ID NO: 22039069**	**Member 4 ID NO: 22014690**
Participated in group discussion	Yes	Yes	Yes	Yes
Helped to monitor group progress	Yes	Yes	Yes	Yes
Contributed useful ideas	Yes	Yes	Yes	Yes
Completed work on time	Yes	Yes	Yes	Yes
Submitted good quality of work	Yes	Yes	Yes	Yes
Communicated effectively with group members	Yes	Yes	Yes	Yes
Helped others with their work when needed	Yes	Yes	Yes	Yes

Good relationship with group members	Yes	Yes	Yes	Yes
Overall was a valuable member of the team	Yes	Yes	Yes	Yes

Self-Reflection *(what I did best and what I should improve in future):*

Throughout the process of working on the assignment, I have cooperated with my teammates, joined in multiple consultations and made changes to my work accordingly to adjust to the group's requirements. Overall, I should improve on participating in more group conversation and providing more ideas to the team.